

CE 773: Geometric Design and Analysis of High-Speed Roadways

Course Project

Geometric Design and Analysis of the Simaria–Matihani Greenfield Bypass Road, Begusarai District, Bihar



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Abstract

Begusarai district in Bihar hosts three major industrial assets, namely the IOCL Barauni Refinery, the NTPC Barauni Thermal Power Station, and the HURL Barauni Fertilizer Plant, yet its only regional arterial, NH-31, suffers chronic congestion, a low average speed of 18 km/h against an 80 km/h design speed, and 142 recorded fatal accidents in 2024. This report presents the geometric design of a proposed greenfield bypass connecting Simaria, Shihma village, and Matihani, intended to divert industrial and through traffic away from the congested urban core of Begusarai. The original scope of the project called for three candidate alignments, but owing to the time available for this coursework exercise, full geometric design was carried through to completion for two of them: a 4-lane, 100 km/h alignment (Alignment 1) and a 2-lane, 80 km/h alignment (Alignment 3) routed to minimise land acquisition through the Makor habitation area. A third, 4-lane, 80 km/h option (Alignment 2) was explored at the routing and superelevation stage in AutoCAD Civil 3D using a Digital Elevation Model (DEM) of the corridor, but was not carried through to full corridor modelling and earthwork quantification. For the two completed alignments, horizontal alignments, spiral transitions, superelevation, vertical profiles, and corridor cross-sections were generated in accordance with IRC:73-1980 and IRC:SP:84 guidelines, and cut-fill earthwork volumes were quantified along each alignment. The resulting designs demonstrate a terrain-responsive, standards-compliant bypass geometry that supports the stated policy objective of reducing NH-31 traffic volume by 38–42% while improving industrial and agricultural connectivity in the region.

Keywords: Highway Geometric Design; Greenfield Bypass; AutoCAD Civil 3D; Horizontal and Vertical Alignment; Superelevation; IRC:73-1980; Earthwork Analysis

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1 Introduction and Study Area

1.1 Project Background

The Simaria to Lakhminia bypass is part of the Government of Bihar's greenfield initiative to reduce traffic congestion and improve regional connectivity around Begusarai. The proposed alignment begins at Simaria, passes through Bind Toli, Shihma, and Matihani, and terminates at Lakhminia. As a greenfield project, the bypass is to be constructed on new land to accommodate rising traffic demand rather than widening the existing corridor.

Alignment definition:

- **Start point:** Simaria (25.44816°N, 85.92636°E)
- **End point:** Matihani (25.421°N, 86.075°E)
- **Intermediate node:** Shihma village (25.433°N, 86.012°E)
- **Total length:** Between 10–20 km (Phase I: Simaria–Shihma–Matihani)
- **Future expansion:** A Phase II extension to Ballia (25.387°N, 86.131°E) has also been proposed.

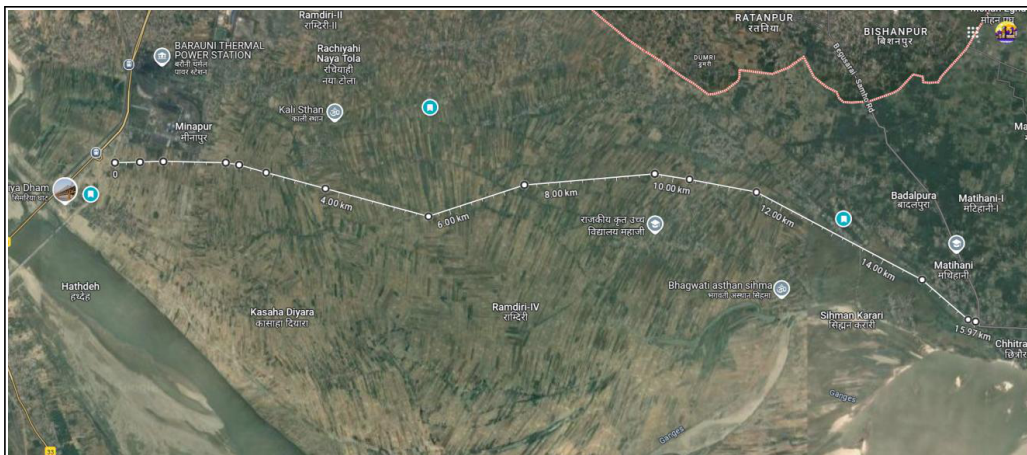


Figure 1: Satellite overview of the Simaria–Shihma–Matihani bypass corridor.

According to the preliminary project brief, the bypass is expected to decongest NH-31 by diverting 40% of commercial traffic away from Begusarai's urban core while connecting critical industrial and agricultural zones. For the purposes of this coursework exercise, the actual Bihar government proposal (the Simaria–Ballia Bypass Road) has been broken into two design phases, of which this report covers Phase I.

1.2 Begusarai: An Emerging Urban-Industrial Hub

Begusarai district, sometimes described as the “Industrial Capital of Bihar,” hosts three pillars of the regional economy:

1. **Indian Oil Corporation Limited (IOCL) Barauni Refinery**, with a capacity of 6 million metric tonnes/year, generating 35% of Bihar’s petroleum products.
2. **Barauni Thermal Power Station (NTPC)**, with an installed capacity of 720 MW, supplying power to 12 districts.
3. **Barauni Fertilizer Plant (HURL)**, with an annual production of 3.85 lakh metric tonnes of urea.

The district’s population is estimated at 1.9 million (2025), growing at a CAGR of 2.8% (2011–2025) against Bihar’s state-wide average of 2.3%. Simaria itself sits within the Barauni Industrial Complex, on predominantly flat terrain well suited to alignment design, but the area is prone to seasonal flooding owing to its proximity to the Ganga river.

1.3 Existing Transportation Network and Challenges

Table 1 summarises the key routes currently serving the study area.

Table 1: Key existing routes in the Begusarai study area (2025 condition).

Route	Type	Condition	Peak-hour traffic (PCU/hr)
NH-31 (Urban)	4-lane	Poor (PCI: 2.1/5)	4,200
SH-78	2-lane	Fair (PCI: 3.4/5)	1,850
Barauni–Begusarai Road	2-lane	Critical	3,100

Identified problems:

- **Chronic congestion:** Daily traffic jams at Kapasia Chowk (NH-31/SH-78 junction); average vehicle speed of 18 km/h against a design speed of 80 km/h.
- **Safety issues:** 142 fatal accidents recorded in 2024 on the NH-31 Begusarai stretch, with black spots at Khatopur Chowk (23 accidents) and IOCL Gate No. 2 (17 accidents).
- **Connectivity gaps:** No direct link between the industrial zone (Simaria) and agricultural markets (Matihani).

Two interventions are already under way, namely the 4.3 km Begusarai Elevated Road (Kapasia Chowk to Jail Chowk, completion expected December 2025) and a proposed Patna–Purnia Expressway link to the Bakhri Industrial Park, but neither addresses regional connectivity in the Simaria area, and unmet needs remain for last-mile connectivity to Shihma’s agro-processing units and emergency access for NTPC’s coal transportation.

1.4 Project Objectives and Expected Outcomes

Primary objectives:

1. Reduce NH-31 traffic volume by 38–42% through diversion.
2. Cut travel time on the Simaria–Ballia corridor from 82 minutes to 32 minutes.
3. Enable 24/7 movement of oversized cargo for IOCL/NTPC.
4. Connect currently unserved villages to the district headquarters.

Anticipated benefits (per the proposed project brief and supporting articles):

- *Economic*: an estimated Rs. 2,300 crore/year in logistics-cost savings (FICCI 2025 estimate), and a 15% increase in fertilizer-plant output through more reliable coal supply.
- *Social*: a projected 45% reduction in road fatalities, and 22 new schools brought within a 30-minute access radius.
- *Environmental*: an estimated 18,000 tonnes CO₂/year reduction via shorter routes, and 32 ha of green belt along the bypass.

1.5 Villages Along the Route

The Simaria–Shihma bypass alignment passes through several key settlements, including Shihma (pincode 851129, Matihani block) and Malhipur (pincode 851118, under the Barauni Panchayat Samiti). These villages stand to benefit directly from improved market access, reduced travel times, and easier transportation of both agricultural produce and industrial goods.

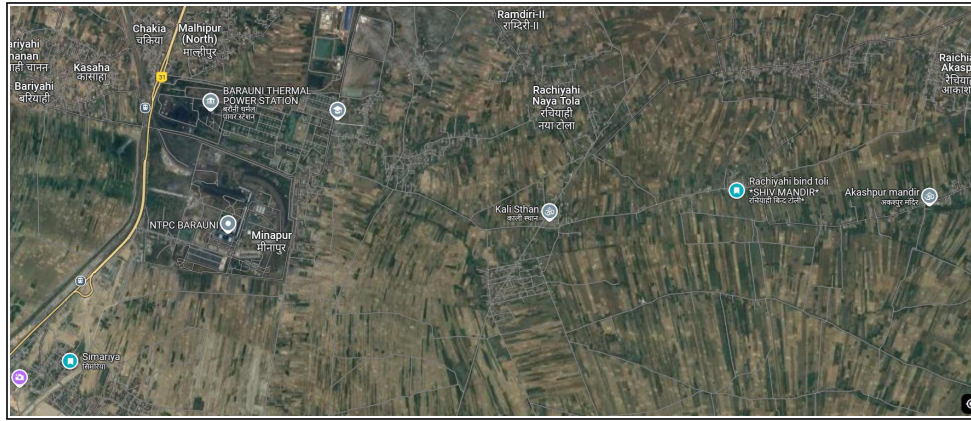


Figure 2: Blue circle: IOCL Barauni industrial complex; red circles: villages affected by the alignment.

2 Methodology (Design Process)

2.1 Design Criteria

All candidate alignments considered, whether completed in full or explored preliminarily, are bypass roads intended to divert traffic away from Begusarai’s urban core.

- **Design speed:** Alignment 1 is designed for 100 km/h, reflecting higher-speed industrial traffic flow; Alignments 2 and 3 are designed for 80 km/h, suitable for general bypass traffic.
- **Lane configuration:** Alignments 1 and 2 are 4-lane roads, sized for expected industrial vehicle volumes. Alignment 3 is a 2-lane road, since it passes through the Makor habitation area where land acquisition and environmental constraints necessitate a narrower cross-section.
- **Geometric design standards:** Both completed alignments follow IRC:73-1980 for horizontal and vertical curves, superelevation, and cross-section profile. Curve radii scale with design speed (larger radii for the 100 km/h alignment than for the 80 km/h alignment), and superelevation is computed from design speed and curve radius subject to a maximum rate of 7%, per IRC:SP:84 for plain terrain.
- **Spiral length:** Spiral transitions of 70 to 90 m are used to provide smooth transitions between tangents and curves.
- **Traffic considerations:** Mixed traffic, comprising industrial vehicles, local traffic, and through traffic bypassing Begusarai, informs lane configuration and curve radii; heavier vehicles require wider curves and stronger superelevation for safety.

2.2 Data Collection

A Digital Elevation Model (DEM) provided the terrain data used for alignment creation in AutoCAD Civil 3D. The DEM was used to (i) generate terrain surfaces for accurate alignment modelling, (ii) analyse elevation profiles and identify flood-prone areas near the Ganga river, and (iii) ensure the proposed alignment respects natural landforms such as rivers, hillocks, and floodplains. In the absence of direct traffic-count data for this coursework exercise, the DEM was the primary source of terrain and elevation information guiding alignment placement.

2.3 Civil 3D Model Development

The alignment, profile, and cross-sections for Alignments 1 and 3 were developed in full in AutoCAD Civil 3D following a standard corridor-design workflow; Alignment 2 was taken only as far as the horizontal alignment and superelevation stages before design effort was concentrated on the two alignments carried to completion:

DEM import → Horizontal alignment → Superelevation & widening → Vertical alignment → Assembly
→ Corridor → Volume/earthwork calculation

3 Alignment 1: Technical Details and Design Analysis

Alignment 1 was designed to minimise travel time relative to the other alignments while keeping cut-and-fill earthwork volumes low.

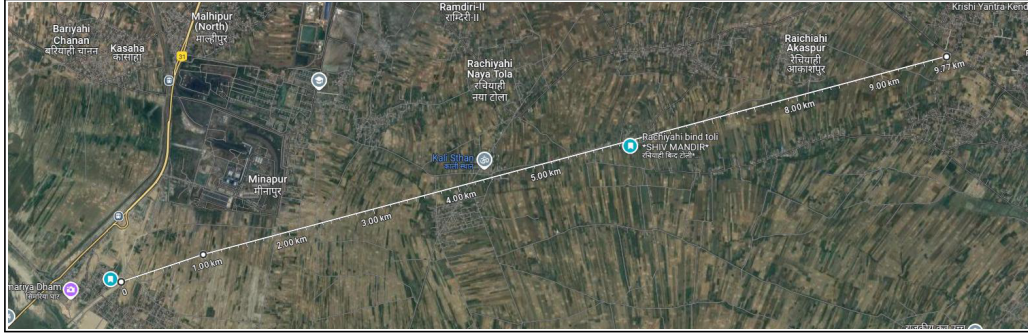


Figure 3: Alignment 1: approximate origin-destination routing (Simaria, NH-31, to the undivided carriageway road near Matihani).

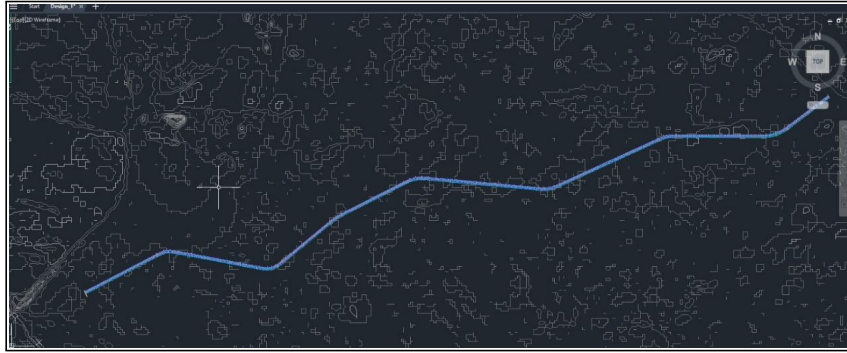


Figure 4: Horizontal alignment of Alignment 1 in AutoCAD Civil 3D.

Table 2: Technical parameters of Alignment 1.

Parameter	Value
Design speed	100 km/h
Lane configuration	4-lane divided
Superelevation rate	6%
Total curves	7
Alignment length	9.980 km
Curve radius	250 m
Spiral length	90 m
Minimum spiral length	45 m
Spiral transition type	Spiral-in / Spiral-out
Start station	0+000.0
End station	9+984.0
Elevation at start	18.75 m
Elevation at end	17.50 m

3.1 Detailed Analysis of Design Variables

Table 3 lists the full geometric-element table generated by Civil 3D for Alignment 1: each tangent (line) and spiral-curve-spiral group, its length, spiral parameters, radius, and degree of curvature. Line elements are unconstrained (fixed, defined by two points); spiral-curve-spiral elements are constrained on both sides (free), parameterised as spiral-in, radius, spiral-out. The design-speed field in the Civil 3D template records 80 km/h throughout this table, reflecting the software’s default alignment-creation parameter rather than the 100 km/h target design speed discussed for Alignment 1 elsewhere in this report.

Table 3: Geometric design data for Alignment 1 (tangent/spiral/curve elements, as generated in Civil 3D).

No.	Element	Length (m)	Min. spi- ral (m)	Radius (m)	Min. radius (m)	A	Degree of curv.
1	Line	1003.743	–	–	–	–	–
2.1	Spiral-in	90.000	90.000	–	–	150.000	–
2.2	Curve	75.699	–	250.000	250.000	–	6.9855
2.3	Spiral- out	90.000	90.000	–	–	150.000	–
3	Line	1037.481	–	–	–	–	–
4.1	Spiral-in	100.000	90.000	–	–	158.114	–
4.2	Curve	118.393	–	250.000	250.000	–	6.9855
4.3	Spiral- out	100.000	90.000	–	–	158.114	–
5	Line	769.374	–	–	–	–	–
6.1	Spiral-in	60.000	45.000	–	–	173.205	–
6.2	Curve	49.689	–	500.000	250.000	–	3.4928
6.3	Spiral- out	60.000	45.000	–	–	173.205	–
7	Line	862.321	–	–	–	–	–
8.1	Spiral-in	80.000	55.000	–	–	178.885	–
8.2	Curve	144.674	–	400.000	250.000	–	4.3659
8.3	Spiral- out	80.000	55.000	–	–	178.885	–
9	Line	1368.077	–	–	–	–	–
10.1	Spiral-in	70.000	55.000	–	–	167.332	–
10.2	Curve	138.469	–	400.000	250.000	–	4.3659
10.3	Spiral- out	70.000	55.000	–	–	167.332	–
11	Line	1338.740	–	–	–	–	–
12.1	Spiral-in	80.000	60.000	–	–	167.332	–

No.	Element	Length (m)	Min. spi- ral (m)	Radius (m)	Min. radius (m)	A	Degree of curv.
12.2	Curve	69.991	—	350.000	250.000	—	4.9896
12.3	Spiral- out	80.000	60.000	—	—	167.332	—
13	Line	1042.366	—	—	—	—	—
14.1	Spiral-in	80.000	45.000	—	—	200.000	—
14.2	Curve	242.831	—	500.000	250.000	—	3.4928
14.3	Spiral- out	80.000	45.000	—	—	200.000	—
15	Line	602.215	—	—	—	—	—

Superelevation. For the 100 km/h design speed, superelevation is calculated at a 6% rate and applied symmetrically on both lanes of the 4-lane divided cross-section.

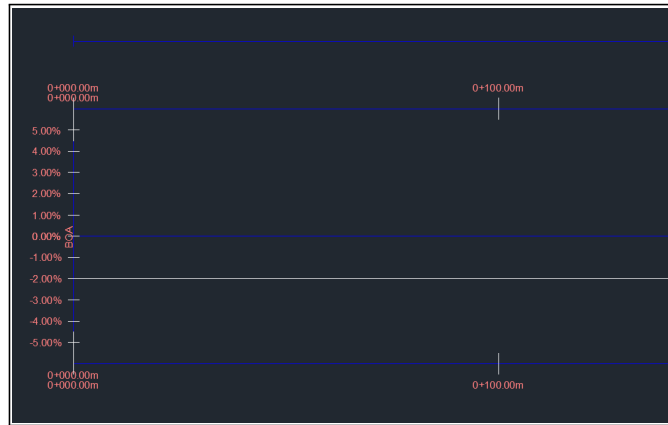


Figure 5: Superelevation transition for Curve 1 of Alignment 1.

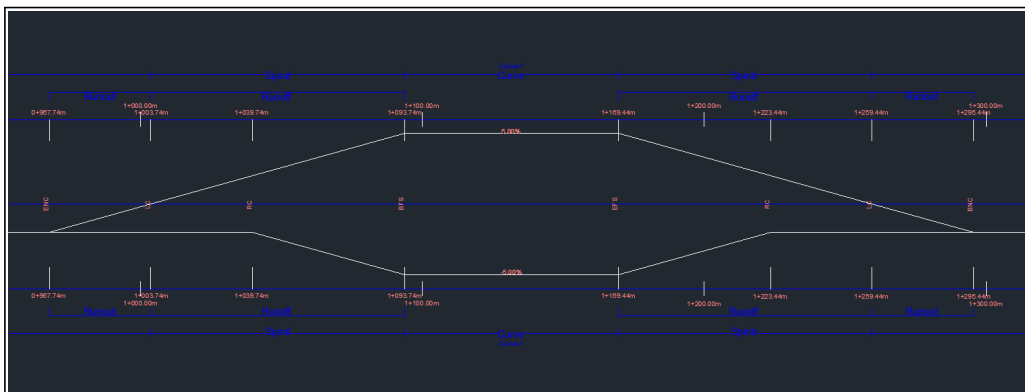


Figure 6: Full superelevation diagram along Alignment 1, showing runoff and runout zones either side of each curve.

Curve radius. The 250 m curve radius adopted for Alignment 1 provides a smooth turning

path suitable for the 100 km/h design speed and is within acceptable limits for high-speed highway design.

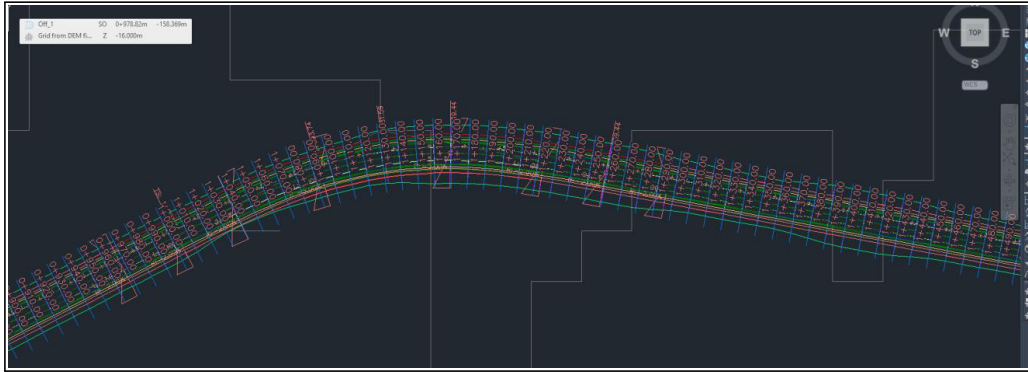


Figure 7: Corridor model generated along a representative curve of Alignment 1.

Spiral length. A 90 m spiral length provides a smooth tangent-to-curve transition, reducing the rate of change of lateral acceleration (jerk) felt by drivers, particularly in high-speed zones; sharper transitions use extended spiral lengths to avoid abrupt superelevation changes.

3.2 Geometric Design Evaluation

Horizontal alignment. The horizontal alignment follows the natural terrain to minimise land acquisition, with curves spaced to allow heavy industrial and mixed traffic to flow without conflict.

Vertical alignment. Vertical grades meet IRC guidelines, keeping the maximum gradient within 4% for high-speed roads and avoiding sharp elevation changes in favour of a smooth longitudinal profile.

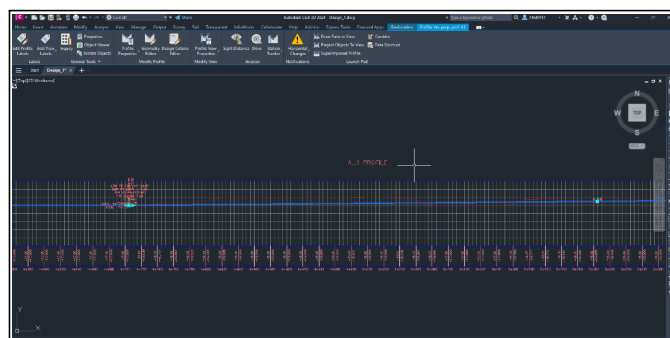


Figure 8: Vertical alignment profile for Alignment 1 in Civil 3D.

Cross-section design. The 4-lane cross-section provides 3.5 m lane widths (the Indian highway standard) and a 2.5 m shoulder for emergency stops and service vehicles, giving adequate width for mixed light-vehicle and heavy-truck traffic without congestion.

3.3 Cut-and-Fill Analysis

Earthwork quantities were computed from the DEM-derived terrain surface. The alignment required elevated (embanked) sections near Bind Toli to account for flood-prone areas close to the Ganga river. Table 4 shows a representative excerpt of the station-by-station volume report generated in Civil 3D; the cumulative net earthwork volume at the final station (9+980.000) was approximately 129,634 m³.

Table 4: Excerpt of the cut/fill volume report for Alignment 1 (Start Sta. 0+010, End Sta. 9+980). All areas in m², all volumes in m³.

Station	Cut area	Cut vol.	Fill area	Fill vol.	Cum. net vol.
0+010.000	10.78	0.00	0.00	0.00	0.00
0+050.000	10.78	107.76	0.00	0.00	538.82
0+100.000	10.78	107.76	0.00	0.00	969.87
0+180.000	3.30	49.60	6.67	44.77	1689.55
9+800.000	27.89	256.90	0.00	0.00	126027.90
9+900.000	20.40	203.08	0.00	0.00	128773.17
9+980.000	9.06	85.30	0.54	7.08	129633.83

4 Alignment 2: Preliminary Exploration (Not Carried Forward)

Alignment 2 is similar to Alignment 1 in origin, destination, and 4-lane cross-section, differing primarily in total length as a result of a slightly different routing through the terrain. This option was explored preliminarily in Civil 3D, with horizontal routing and a superelevation diagram generated as shown below, but given the time available for this coursework project, detailed corridor modelling and earthwork quantification were not carried out for this alignment. Design effort was instead concentrated on Alignments 1 and 3, which are presented in full in this report.

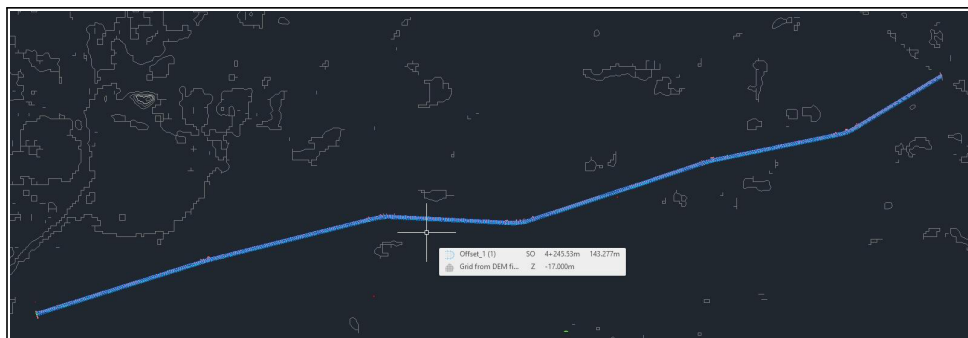


Figure 9: Alignment 2 routing over the DEM-derived terrain grid.

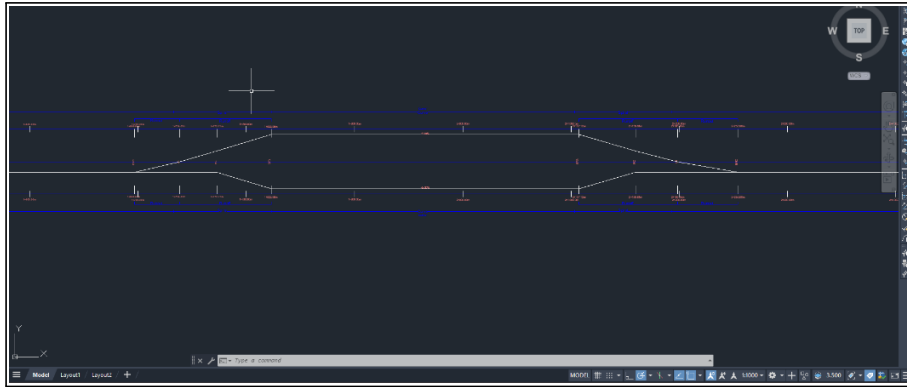


Figure 10: Superelevation diagram for Alignment 2.

5 Alignment 3: 2-Lane Village Connector

Alignment 3 was designed as a 2-lane, 80 km/h bypass connecting Bind Toli and other scattered villages before merging with the existing undivided carriageway. Because it passes through the Makor habitation area, its narrower 2-lane cross-section reflects land-acquisition and environmental constraints rather than a lower expected traffic volume.

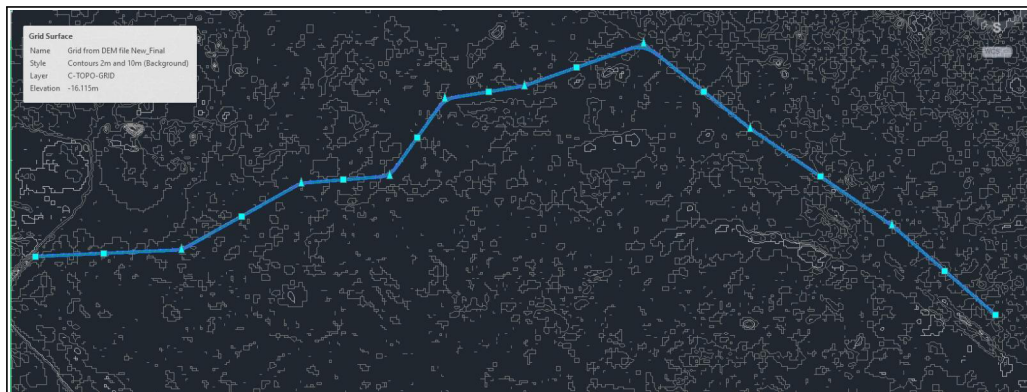


Figure 11: Alignment 3 routing, connecting village areas before widening at the final curve.

Table 5: Design variables for Alignment 3.

Parameter	Value
Design speed	80 km/h
Alignment length	15.66 km
Lane configuration	2-lane road
Superelevation rate	4–5% (depending on the curve)
Curve radius	200 m to 500 m
Total curves	9
Spiral length	45 m – 90 m
Minimum spiral length	45 m
Start station	0+000.0
End station	15+000.0
Spiral type	Spiral-in / Spiral-out
Earthwork	Cut and fill based on DEM data

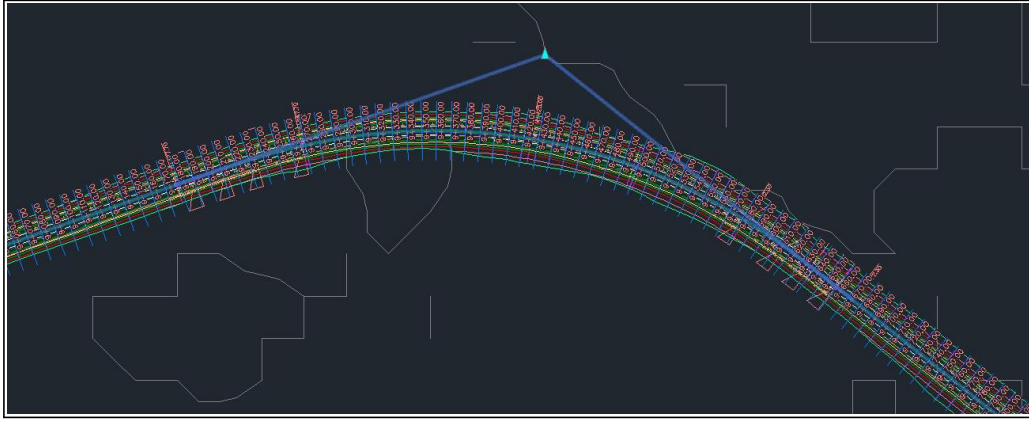


Figure 12: Corridor model along a representative curve of Alignment 3.

The cut-and-fill volume report for Alignment 3 (Start Sta. 0+010, End Sta. 15+660) shows a cumulative net earthwork volume of approximately 293,198 m³ at the final station, reflecting the longer alignment length and the more undulating terrain traversed relative to Alignment 1.

6 Discussion

The two completed alignments illustrate a set of trade-offs typical of greenfield bypass design. Alignment 1, designed for 100 km/h with 250 m curve radii, minimises travel time for industrial and through traffic at the cost of a wider 4-lane cross-section and larger land take. Alignment 3 trades capacity (2 lanes instead of 4) and a lower nominal design speed for a routing that minimises impact on the Makor habitation area, at the expense of a longer overall length (15.66 km versus 9.98 km for Alignment 1) and correspondingly larger cumulative earthwork volumes. A third option, Alignment 2, offering similar 4-lane capacity to Alignment 1 at a more conservative 80 km/h design speed, was explored at the routing and superelevation stage but was not carried through to full corridor design and earthwork quantification within the time

available for this project. For both completed alignments, adherence to IRC:73-1980 curve-radius and superelevation limits, together with DEM-informed vertical design, ensures the alignments remain constructible on the flood-prone, low-relief terrain characteristic of the Simaria–Matihani corridor near the Ganga river.

7 Conclusion

This project successfully achieved:

- Terrain modelling using real DEM/satellite data;
- Engineering-grade horizontal alignment creation for two fully designed candidate bypass routes (Alignments 1 and 3), with a third option (Alignment 2) explored preliminarily;
- Compliance with Indian geometric design standards (IRC:73-1980, IRC:SP:84);
- A policy-aligned justification for bypass development, linking the design directly to the stated regional objectives of decongesting NH-31 and improving industrial and agricultural connectivity around Begusarai.

This design work lays the groundwork for the subsequent stages of a fully implementable bypass design, namely refined vertical profiling, detailed superelevation and widening design, corridor modelling, and earthwork-based cost estimation, all of which were carried out for Alignments 1 and 3 in this report.

References

- [1] Indian Roads Congress. *IRC:73-1980 – Geometric Design Standards for Rural (Non-Urban) Highways*.
- [2] Indian Roads Congress. *IRC:SP:84 – Manual of Specifications and Standards for Six Lining of Highways*.